

Table of Physical Constants

	Symbol	Value	Units
Planck's constant	h	6.626×10^{-34}	J-s
Speed of light	c	3.0×10^8	m/s
Electronic charge	e (or q)	1.6×10^{-19}	C
Boltzmann's constant	k_B (or k)	1.38×10^{-23}	J/K
Free electron rest mass	m_0	9.1×10^{-31}	kg
Proton rest mass	m_p	1.67×10^{-27}	kg
Avogadro's number	N_A	6.02×10^{23}	mol ⁻¹
Permeability of free space	μ_0	$4\pi \times 10^{-7}$	H/m
Permittivity of free space	ϵ_0	8.85×10^{-12}	F/m
Rydberg constant	R_d	1.097×10^7	m ⁻¹
Bohr radius	a_0	5.292×10^{-11}	m
Gas constant	R	8.31	J/mol-K
Electron-volt	1 eV	1.6×10^{-19}	J
Thermal voltage ($T= 300\text{K}$)	kT/q	0.0259	V